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Radio-frequency circuit and communication device

Abstract

A radio-frequency circuit includes an amplifier circuit, a bias circuit, a bias control circuit, a comparing section, a signal input terminal, an antenna terminal, an attenuation circuit, and a control unit. The amplifier circuit includes a specific transistor. The bias circuit supplies a bias current or a bias voltage to the input terminal of the specific transistor. The bias control circuit supplies a control current or a control voltage to the bias circuit. The comparing section compares a threshold voltage with a power supply voltage of a power supply terminal connected to the output terminal of the specific transistor. The attenuation circuit is connected in a signal path between the signal input terminal and the antenna terminal and is capable of attenuating the radio-frequency signal. The control unit changes an attenuation of the attenuation circuit in accordance with a compared result of the comparing section.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This is a continuation of International Application No. PCT/JP2021/016502 filed on Apr. 23, 2021 which claims priority from Japanese Patent Application No. 2020-106479 filed on Jun. 19, 2020. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND ART

Technical Field

- (1) The present disclosure generally relates to a radio-frequency circuit and a communication device and, more specifically, to a radio-frequency circuit including an amplifier circuit, and a communication device.
- (2) Hitherto, a power amplifier module that includes a power amplifier circuit and a bias circuit is known (see, for example, Patent Document 1).
- (3) The amplifier circuit includes a plurality of amplifiers connected in multiple stages. Each of the amplifiers includes a common-emitter transistor. The collector of the transistor is connected to a power supply voltage through an inductance element.
- (4) The bias circuit includes a plurality of emitter follower transistors and a control IC. The control IC includes a first current source and a second current source. The first current source supplies the collector of each of the plurality of emitter follower transistors with a control current that changes following a change in control voltage. The first current source limits the control current to less than or equal to an upper limit. The second current source supplies the base of each of the plurality of emitter follower transistors with a constant current. Patent Document 1: Japanese Unexamined Patent Application Publication No. 2018-152714

BRIEF SUMMARY

- (5) In the power amplifier module described in Patent Document 1, when the power supply voltage is higher than a rated power supply voltage, there have been variations in bias current or bias voltage supplied to the base of, for example, the transistor of the amplifier because of the influence of variations in h_{fe} (output short-circuit current gain) of the emitter follower **30**. Therefore, with the power amplifier module described in Patent Document 1, the characteristics of at least one of the amplifier circuit and an electronic component to which a radio-frequency signal amplified by the amplifier circuit is input sometimes degrade.
- (6) The present disclosure provides a radio-frequency circuit and a communication device capable of further accurately limiting the output power of an amplifier circuit.
- (7) A radio-frequency circuit according to an aspect of the present disclosure includes an amplifier circuit, a bias circuit, a bias control circuit, a comparing section, a signal input terminal, an antenna terminal, an attenuation circuit, and a control unit. The amplifier circuit includes a specific transistor. The specific transistor has an input terminal and an output terminal. The specific transistor amplifies a radio-frequency signal input to the input terminal and outputs the radio-frequency signal from the output terminal. The bias circuit supplies a bias current or a bias voltage to the input terminal of the specific transistor. The bias control circuit supplies a control current or a control voltage to the bias circuit. The comparing section compares a threshold voltage with a power supply voltage of a power supply terminal connected to the output terminal of the specific transistor. The signal input terminal is connected to the input terminal of the specific transistor, and the radio-frequency signal is input to the signal input terminal. The attenuation circuit is connected

in a signal path between the signal input terminal and the antenna terminal and is capable of attenuating the radio-frequency signal. The control unit changes an attenuation of the attenuation circuit in accordance with a compared result off the comparing section.

(8) A communication device according to an aspect of the present disclosure includes the radio-frequency circuit and a signal processing circuit. The signal processing circuit is connected to the radio-frequency circuit.

(9) With the radio-frequency circuit and the communication device according to the above-described aspects of the present disclosure, it is possible to further accurately limit the output power of the amplifier circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a circuit diagram of a radio-frequency circuit and a communication device according to a first embodiment.

(2) FIG. 2 is a circuit diagram of a power amplifier circuit of the above-described radio-frequency circuit.

(3) FIG. 3A is a circuit diagram of an attenuation circuit of the above-described radio-frequency circuit.

(4) FIG. 3B is a circuit diagram of an attenuation circuit of a radio-frequency circuit according to a first modification of the first embodiment. FIG. 3C is a circuit diagram of an attenuation circuit of a radio-frequency circuit according to a second modification of the first embodiment. FIG. 3D is a circuit diagram of an attenuation circuit of a radio-frequency circuit according to a third modification of the first embodiment. FIG. 3E is a circuit diagram of an attenuation circuit of a radio-frequency circuit according to a fourth modification of the first embodiment.

(5) FIG. 4A is a circuit diagram of an attenuation circuit of a radio-frequency circuit according to a fifth modification of the first embodiment. FIG. 4B is a circuit diagram of an attenuation circuit of a radio-frequency circuit according to a sixth modification of the first embodiment. FIG. 4C is a circuit diagram of an attenuation circuit of a radio-frequency circuit according to a seventh modification of the first embodiment.

(6) FIG. 5 is a circuit diagram of a radio-frequency circuit and a communication device according to a second embodiment.

(7) FIG. 6 is a circuit diagram of a radio-frequency circuit and a communication device according to a third embodiment.

(8) FIG. 7 is a circuit diagram of a radio-frequency circuit and a communication device according to a fourth embodiment.

DETAILED DESCRIPTION

First Embodiment

(9) Hereinafter, a radio-frequency circuit **100** and a communication device **300** according to a first embodiment will be described with reference to FIGS. 1 to 4C.

(10) (1) Radio-Frequency Circuit and Communication Device

(11) (1.1) Circuit Configuration of Radio-Frequency Circuit and Communication Device

(12) The radio-frequency circuit **100** according to the first embodiment is used in, for example, the communication device **300**. The communication device **300** is, for example, a mobile phone (for example, a smartphone); however, the configuration is not limited thereto. The communication device **300** may be, for example, a wearable terminal (for example, a smart watch). The radio-frequency circuit **100** is, for example, a circuit that supports a 4G (fourth generation mobile communication) standard, a 5G (fifth generation mobile communication) standard, or the like. The 4G standard is, for example, a 3GPP long term evolution (LTE) standard. The 5G standard is, for

example, 5G new ratio (NR). The radio-frequency circuit **100** may be a circuit that supports carrier aggregation and dual connectivity.

(13) The radio-frequency circuit **100** is, for example, configured to be capable of amplifying a transmission signal (radio-frequency signal) input from the signal processing circuit **301** and outputting the transmission signal to an antenna **310**. The signal processing circuit **301** is not a component element of the radio-frequency circuit **100** but a component element of the communication device **300** that includes the radio-frequency circuit **100**. The radio-frequency circuit **100** is, for example, controlled by the signal processing circuit **301** of the communication device **300**. The communication device **300** includes the radio-frequency circuit **100** and the signal processing circuit **301**. The communication device **300** further includes the antenna **310**.

(14) The signal processing circuit **301** includes, for example, an RF signal processing circuit **302** and a baseband signal processing circuit **303**. The RF signal processing circuit **302** is, for example, a radio frequency integrated circuit (RFIC), and performs signal processing on a radio-frequency signal. The RF signal processing circuit **302**, for example, performs signal processing on a radio-frequency signal (transmission signal) output from the baseband signal processing circuit **303** by up conversion, or the like and outputs the processed radio-frequency signal. The baseband signal processing circuit **303** is, for example, a baseband integrated circuit (BBIC). The baseband signal processing circuit **303** generates an I-phase signal and a Q-phase signal from a baseband signal. The baseband signal is, for example, an audio signal, an image signal, or the like input from the outside. The baseband signal processing circuit **303** performs IQ modulation process by synthesizing the I-phase signal with the Q-phase signal and outputs a transmission signal. At this time, the transmission signal is generated as a modulation signal (IQ signal) obtained by modulating the amplitude of a carrier wave signal with a predetermined frequency at a period longer than the period of the carrier wave signal. The radio-frequency circuit **100** transmits a radio-frequency signal (transmission signal) between the antenna **310** and the RF signal processing circuit **302** of the signal processing circuit **301**.

(15) The radio-frequency circuit **100** includes a power amplifier circuit **1**. The radio-frequency circuit **100** further includes an output matching circuit **101**, a first switch **102**, a filter **103**, a second switch **104**, and an antenna terminal **105**.

(16) The power amplifier circuit **1** includes an amplifier circuit **2**. The amplifier circuit **2**, for example, amplifies an input signal input from the signal processing circuit **301** via a signal input terminal **13** of the radio-frequency circuit **100** and outputs the input signal. The input signal is a radio-frequency signal (transmission signal) in a predetermined frequency band. The predetermined frequency band includes, for example, a plurality of communication bands different from each other. The amplifier circuit **2** has a plurality of (two in the illustrated example) amplifiers **20**.

(17) Each of the plurality of amplifiers **20** includes an amplifier transistor Tr2 (see FIG. 2). Each of the plurality of transistors Tr2 has an input terminal and an output terminal. Each of the plurality of transistors Tr2 has an input/output common terminal. Each of the plurality of transistors Tr2 is, for example, a bipolar transistor. In this case, the input terminal, output terminal, and common terminal of the transistor Tr2 are respectively the base, collector, and emitter of the bipolar transistor. In the amplifier circuit **2**, the common terminal (emitter) of each transistor Tr2 is connected to a ground (grounded).

(18) In the amplifier circuit **2**, the plurality of amplifiers **20** is connected in multiple stages. Therefore, in the amplifier circuit **2**, the plurality of transistors Tr2 is connected in multiple stages. Hereinafter, for the sake of convenience of description, of the plurality of amplifiers **20**, the amplifier **20** in the first stage (driver stage) is referred to as first-stage amplifier **21**, and the amplifier **20** in the final stage is referred to as final-stage amplifier **22**. Of the plurality of transistors Tr2, the transistor Tr2 in the first stage (driver stage) is referred to as first-stage transistor Tr21, and the transistor Tr2 in the final stage (output stage) is referred to as final-stage transistor Tr22.

(19) The amplifier circuit **2** further includes a matching circuit **23** provided between an input terminal **27** and the first-stage amplifier **21**, and a matching circuit **24** provided between the first-stage amplifier **21** and the final-stage amplifier **22**. The matching circuit **23** is a circuit for matching the impedance between the first-stage amplifier **21** and the signal processing circuit **301**. The matching circuit **24** is a circuit (interstage matching circuit) for matching the impedance between the first-stage amplifier **21** and the final-stage amplifier **22**.

(20) The output matching circuit **101** is provided in a signal path between the amplifier circuit **2** and the first switch **102**. The output matching circuit **101** is a circuit for matching the impedance between the amplifier circuit **2** and the filter **103**. The output matching circuit **101** is made up of, for example, one inductor; however, the configuration is not limited thereto. The output matching circuit **101** may include, for example, a plurality of inductors and a plurality of capacitors.

(21) The first switch **102** is provided between the output matching circuit **101** and the filter **103**. The first switch **102** has a common terminal and a plurality of selection terminals. The common terminal of the first switch **102** is connected to the amplifier circuit **2** via the output matching circuit **101**. One of the plurality of selection terminals of the first switch **102** is connected to the filter **103**. The first switch **102** is, for example, a switch capable of connecting at least one of the plurality of selection terminals to the common terminal. Here, the first switch **102** is, for example, a switch capable of one-to-one connection and one-to-multiple connection. The first switch **102** is a switch capable of switching between signal paths for a plurality of transmission signals in communication bands different from each other. The first switch **102**, for example, switches a connection status between the common terminal and the plurality of selection terminals in accordance with a control signal input from the signal processing circuit **301**. The first switch **102** is, for example, a switch integrated circuit (IC).

(22) The filter **103** is a filter that has a transmission band of at least one communication band (for example, Band **3**) of the above-described plurality of communication bands as a pass band. The filter **103** is, for example, a one-chip acoustic wave filter. In the filter **103**, each of a plurality of series arm resonators and a plurality of parallel arm resonators is made up of an acoustic wave resonator. The acoustic wave filter is, for example, a surface acoustic wave filter that uses surface acoustic waves. In the surface acoustic wave filter, each of the plurality of series arm resonators and the plurality of parallel arm resonators is, for example, a surface acoustic wave (SAW) resonator.

(23) The second switch **104** is provided between the filter **103** and the antenna terminal **105**. The second switch **104** is a switch connected to the antenna terminal **105**. The second switch **104** has a common terminal and a plurality of selection terminals. The common terminal of the second switch **104** is connected to the antenna terminal **105**. One of the plurality of selection terminals of the second switch **104** is connected to the filter **103**. The second switch **104**, for example, switches a connection status between the common terminal and the plurality of selection terminals in accordance with a control signal input from the signal processing circuit **301**. The second switch **104** is, for example, a switch IC.

(24) The antenna terminal **105** is connected to the antenna **310**.

(25) In the radio-frequency circuit **100**, a transmission signal output from the amplifier circuit **2** is transmitted to the antenna **310** through the output matching circuit **101**, the first switch **102**, the filter **103**, the second switch **104**, and the antenna terminal **105**.

(26) (1.2) Circuit Configuration of Power Amplifier Circuit

(27) As shown in FIG. **2**, the power amplifier circuit **1** includes the amplifier circuit **2**, a bias circuit **3**, a bias control circuit **54**, a comparing section **6**, an attenuation circuit **4**, and a control unit **55**. The amplifier circuit **2** includes a specific transistor Tr2A (for example, the first-stage transistor Tr2). The specific transistor Tr2A has an input terminal and an output terminal. The specific transistor Tr2A amplifies a radio-frequency signal input to the input terminal and outputs the radio-frequency signal from the output terminal. The bias circuit **3** supplies a bias current to the input

terminal of the specific transistor Tr2A; however, the configuration is not limited thereto. The bias circuit 3 may be configured to supply a bias voltage to the input terminal of the specific transistor Tr2A. The bias control circuit 54 supplies a control current I_{cont} to the bias circuit 3; however, the configuration is not limited thereto. The bias control circuit 54 may be configured to supply a control voltage to the bias circuit 3. The comparing section 6 compares a threshold voltage (reference voltage) V_{ref} with a power supply voltage V_{cc} of a power supply terminal 11 connected to the output terminal of the specific transistor Tr2A. The attenuation circuit 4 is connected in a signal path P1 (see FIG. 1) between the signal input terminal 13 and the antenna terminal 105 and is capable of attenuating a radio-frequency signal. The control unit 55 controls the attenuation circuit 4.

(28) The amplifier circuit 2 includes a plurality of (two in the illustrated example) amplifier transistors Tr2 (see FIG. 2). Each of the plurality of transistors Tr2 has an input terminal and an output terminal. Each of the plurality of transistors Tr2 has an input/output common terminal. Each of the plurality of transistors Tr2 is, for example, a bipolar transistor. In this case, the input terminal, output terminal, and common terminal of the transistor Tr2 are respectively the base, collector, and emitter of the bipolar transistor. In the amplifier circuit 2, the common terminal (emitter) of each transistor Tr2 is connected to a ground (grounded).

(29) In the amplifier circuit 2, the plurality of transistors Tr2 is connected in multiple stages. Hereinafter, for the sake of convenience of description, of the plurality of transistors Tr2, the transistor Tr2 in the first stage (driver stage) is referred to as first-stage transistor Tr21, and the transistor Tr2 in the final stage (output stage) is referred to as final-stage transistor Tr22.

(30) The amplifier circuit 2 has the input terminal 27 and an output terminal 28. In the amplifier circuit 2, the input terminal (base) of the first-stage transistor Tr21 is connected to the input terminal 27 via the matching circuit 23. The output terminal (collector) of the first-stage transistor Tr21 is connected to the power supply terminal 11. The output terminal (collector) of the first-stage transistor Tr21 is connected to the input terminal (base) of the final-stage transistor Tr22 via the matching circuit 24. The common terminal (emitter) of the first-stage transistor Tr21 is grounded. The first-stage transistor Tr2 amplifies an input signal input to the input terminal and outputs the input signal from the output terminal.

(31) The input terminal (base) of the final-stage transistor Tr22 is connected to the output terminal (collector) of the first-stage transistor Tr21 via the matching circuit 24. The output terminal (collector) of the final-stage transistor Tr22 is connected to the power supply terminal 11. The output terminal of the final-stage transistor Tr22 is connected to the output terminal 28 via a matching circuit 25. The common terminal (emitter) of the final-stage transistor Tr22 is grounded.

(32) The amplifier circuit 2 may have the matching circuit 25 provided between the output terminal of the final-stage transistor Tr22 and the output terminal 28 of the amplifier circuit 2.

(33) The amplifier circuit 2 amplifies an input transmission signal with the first-stage transistor Tr21, further amplifies the transmission signal with the final-stage transistor Tr22, and outputs the transmission signal. In other words, the final-stage transistor Tr22 further amplifies the transmission signal amplified by the first-stage transistor Tr21 and outputs the transmission signal.

(34) The bias circuit 3 has two emitter followers 30 in a one-to-one correspondence with the two transistors Tr2. Each of the two emitter followers 30 includes a bipolar transistor Tr3. Each of the two emitter followers 30 supplies a bias current to the base of an associated one of the two transistors Tr2. The two bipolar transistors Tr3 are bias transistors.

(35) The bias circuit 3 has the plurality of (two in the illustrated example) emitter followers 30 in a one-to-one correspondence with the plurality of (two in the illustrated example) transistors Tr2. Hereinafter, regarding the two emitter followers 30, the emitter follower 30 associated with the first-stage transistor Tr21 may be referred to as first emitter follower 31, and the emitter follower 30 associated with the final-stage transistor Tr22 may be referred to as second emitter follower 32. Regarding the two bipolar transistors Tr3, the bipolar transistor Tr3 included in the first emitter

follower **31** may be referred to as first bipolar transistor Tr**31**, and the bipolar transistor Tr**3** included in the second emitter follower **32** may be referred to as second bipolar transistor Tr**32**.

(36) The base of the first bipolar transistor Tr**31** is connected to a current source **52** via a resistance R**31**. The emitter of the second bipolar transistor Tr**31** is connected to the input terminal (base) of the first-stage transistor Tr**21**. The collector of the first bipolar transistor Tr**31** is connected to a battery terminal **12** of the radio-frequency circuit **100**. The base of the second bipolar transistor Tr**32** is connected to the current source **52** included in the bias control circuit **54** via a resistance R**32**. The emitter of the second bipolar transistor Tr**32** is connected to the input terminal (base) of the final-stage transistor Tr**22**. The collector of the second bipolar transistor Tr**32** is connected to the battery terminal **12**. In the bias circuit **3**, a current I_{ce} from the battery terminal **12** is supplied to a junction point between the collectors of the plurality of bipolar transistors Tr**3**.

(37) The bias control circuit **54** includes the current source **52** as described above. The current source **52** supplies the control current I_{cont} to the base of each of the first bipolar transistor Tr**31** and the second bipolar transistor Tr**32**. The control current I_{cont} is a constant current.

(38) The first emitter follower **31** supplies a bias current from the emitter of the first bipolar transistor Tr**31** to the input terminal (base) of the first-stage transistor Tr**21**. The bias current is a bias signal for controlling a bias point (operating point) of the first-stage transistor Tr**21**.

(39) The second emitter follower **32** supplies a bias current from the emitter of the second bipolar transistor Tr**32** to the input terminal (base) of the final-stage transistor Tr**22**. The bias current is a bias signal for controlling a bias point (operating point) of the final-stage transistor Tr**22**.

(40) Two diodes D**311**, D**312** are connected in series between the ground and the base of the first bipolar transistor Tr**31** included in the first emitter follower **31**. Each of the two diodes D**311**, D**312** is configured to connect the base and collector of an npn transistor.

(41) Two diodes D**321**, D**322** are connected in series between the ground and the base of the second bipolar transistor Tr**32** included in the second emitter follower **32**. Each of the two diodes D**321**, D**322** is configured to connect the base and collector of an npn transistor.

(42) The amplifier circuit **2** and the bias circuit **3** are included in, for example, one IC chip. The IC chip including the amplifier circuit **2** and the bias circuit **3** is, for example, a GaAs IC chip. In this case, each of the two transistors Tr**2** is, for example, a heterojunction bipolar transistor (HBT).

(43) The IC chip including the amplifier circuit **2** and the bias circuit **3** is not limited to a GaAs IC chip, and may be, for example, an Si IC chip having the amplifier circuit **2** or an SiGe IC chip having the amplifier circuit **2**.

(44) The bias control circuit **54** supplies the control current I_{cont} to the bias circuit **3**. The radio-frequency circuit **100** has a control circuit **5** including the bias control circuit **54**. The control circuit **5** is, for example, a control IC that controls the amplifier circuit **2** and the bias circuit **3**. When the bias circuit **3** has a plurality of (here, two) emitter followers **30**, the current I_{ce} from the battery terminal **12** is diverted to the plurality of emitter followers **30**. The control current I_{cont} from the bias control circuit **54** is diverted to the bases of the two bipolar transistors Tr**3** of the bias circuit **3**.

(45) The comparing section **6** compares the threshold voltage V_{ref} with the power supply voltage V_{cc} of the power supply terminal **11** connected to the output terminal (collector) of the specific transistor Tr**2A** (here, the first-stage transistor Tr**21**). The power supply terminal **11** is, for example, a terminal connected to a power management integrated circuit (IC) **330** of the communication device **300** and supplied with the power supply voltage V_{cc} from the power management IC **330**.

(46) The comparing section **6** includes a comparator CP**1** that compares a threshold voltage V_{ref} with the power supply voltage V_{cc} of the power supply terminal **11**. The comparator CP**1** has a first input terminal, a second input terminal, and an output terminal. The first input terminal of the comparator CP**1** is connected to a node N**1** between the power supply terminal **11** and the output terminal (collector) of the first-stage transistor Tr**21**. The first input terminal of the comparator CP**1** is connected to the power supply terminal **11** without necessarily intervening a circuit element. Therefore, no circuit element that generates a voltage drop, such as a resistive element, is provided

between the power supply terminal **11** and the node **N1**. Thus, the power supply voltage V_{cc} is applied to the first input terminal of the comparator **CP1**. The threshold voltage V_{ref} is applied from the control unit **55** of the control circuit **5** to the second input terminal of the comparator **CP1**. The output terminal of the comparator **CP1** is connected to the control unit **55**.

(47) In the comparator **CP1**, for example, the first input terminal is a non-inverting input terminal, and the second input terminal is an inverting input terminal. In the comparing section **6**, when the power supply voltage V_{cc} does not exceed the threshold voltage V_{ref} , the signal level (voltage level) of an output signal from the output terminal of the comparator **CP1** is a first level (also referred to as low level). In the comparing section **6**, when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , the signal level of an output signal from the output terminal of the comparator **CP1** is a second level (also referred to as high level) higher than the first level.

(48) The attenuation circuit **4** is connected in the signal path **P1** between the signal input terminal **13** and the antenna terminal **105**. More specifically, the attenuation circuit **4** is connected between the output terminal of the first-stage transistor **Tr21** serving as the specific transistor **Tr2A** and the input terminal of the final-stage transistor **Tr22**.

(49) As shown in FIG. **3A**, the attenuation circuit **4** includes a resistive element **41** and a switch element **42**. The resistance value of the attenuation circuit **4** is changed in accordance with a compared result of the comparing section **6**. The attenuation circuit **4** is, for example, controlled by the control unit **55**. The switch element **42** is, for example, a semiconductor switch element having a control terminal and a pair of main terminals, and the control unit **55** is connected to the control terminal. The semiconductor switch element is, for example, a normally-on metal-oxide-semiconductor field effect transistor (MOSFET).

(50) The control unit **55** increases the resistance value of the attenuation circuit **4** when the power supply voltage V_{cc} is higher than the threshold voltage V_{ref} as compared to when the power supply voltage V_{cc} is lower than the threshold voltage V_{ref} . In the radio-frequency circuit **100**, when the signal level of the output signal of the comparator **CP1** is the first level, the control unit **55** controls the switch element **42** of the attenuation circuit **4** to an on state; whereas, when the signal level of the output signal of the comparator **CP1** is the second level, the control unit **55** controls the switch element **42** of the attenuation circuit **4** to an off state. Therefore, in the attenuation circuit **4**, when the power supply voltage V_{cc} is lower than the threshold voltage V_{ref} , the resistance value of the attenuation circuit **4** is substantially zero (about the on resistance of the switch element **42**); whereas, when the power supply voltage V_{cc} is higher than the threshold voltage V_{ref} , the resistance value of the attenuation circuit **4** is the resistance value of the resistive element **41**. The resistance value of the resistive element **41** is greater than the on resistance of the switch element **42**. The threshold voltage V_{ref} is a voltage higher by a predetermined voltage (for example, greater than or equal to 0.1 V and less than or equal to 0.2 V) than the rated power supply voltage of the power supply voltage V_{cc} .

(51) The control circuit **5** further includes the control unit **55** that controls the attenuation circuit **4** as described above. The control unit **55** controls the attenuation circuit **4** in accordance with a compared result of the comparing section **6**. In other words, the control unit **55** controls the switch element **42** of the attenuation circuit **4** in accordance with an output signal of the comparator **CP1**.

(52) The control circuit **5** and the comparator **CP1** are included in, for example, one IC chip. The IC chip including the control circuit **5** and the comparator **CP1** is, for example, an Si IC chip.

(53) The control circuit **5** is connected to, for example, the signal processing circuit **301**. The control circuit **5** controls the amplifier circuit **2** and the bias circuit **3** in accordance with a control signal acquired from the signal processing circuit **301**. The control circuit **5** controls the amplifier circuit **2** and the bias circuit **3** in accordance with a control signal from the RF signal processing circuit **302** of the signal processing circuit **301**.

(54) (1.3) Operation of Radio-Frequency Circuit

(55) In the radio-frequency circuit **100**, the power amplifier circuit **1**, for example, amplifies a

transmission signal from the signal processing circuit **301** and outputs the transmission signal. In the power amplifier circuit **1**, the amplifier circuit **2** amplifies an input signal that is an input transmission signal in a predetermined frequency band and outputs the input signal.

(56) In the power amplifier circuit **1**, when the power supply voltage V_{cc} of the power supply terminal **11** is the rated power supply voltage, the power supply voltage V_{cc} does not exceed the threshold voltage V_{ref} , and the signal level of the output signal of the comparator **CP1** is the first level. When the output signal of the comparator **CP1** is the first level, the control unit **55** sets the switch element **42** of the attenuation circuit **4** to an on state. Thus, a radio-frequency signal that passes through the attenuation circuit **4** and enters the input terminal of the final-stage transistor **Tr22** is not attenuated.

(57) In the power amplifier circuit **1**, when the power supply voltage V_{cc} of the power supply terminal **11** becomes a voltage higher by a predetermined voltage (for example, greater than or equal to 0.1 V and less than or equal to 0.2 V) or greater than the rated power supply voltage, the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , and the signal level of the output signal of the comparator **CP1** is the second level. When the output signal of the comparator **CP1** is the second level, the control unit **55** sets the switch element **42** of the attenuation circuit **4** to an off state. Thus, a radio-frequency signal that passes through the attenuation circuit **4** and enters the input terminal of the final-stage transistor **Tr22** is attenuated by the attenuation circuit **4**. Thus, in the power amplifier circuit **1**, the output power of the final-stage transistor **Tr22** is limited. In this way, in the power amplifier circuit **1**, the input voltage-output voltage characteristics of the amplifier circuit **2** depend on the magnitude relation between the power supply voltage V_{cc} and the threshold voltage V_{ref} .

(58) (2) Summary

(59) (2.1) Radio-Frequency Circuit

(60) A radio-frequency circuit **100** according to the first embodiment includes an amplifier circuit **2**, a bias circuit **3**, a bias control circuit **54**, a comparing section **6**, a signal input terminal **13**, an antenna terminal **105**, an attenuation circuit **4**, and a control unit **55**. The amplifier circuit **2** includes a specific transistor **Tr2A**. The specific transistor **Tr2A** has an input terminal and an output terminal. The specific transistor **Tr2A** amplifies a radio-frequency signal input to the input terminal and outputs the radio-frequency signal from the output terminal. The bias circuit **3** supplies a bias current I_{cont} or a bias voltage to the input terminal of the specific transistor **Tr2A**. The bias control circuit **54** supplies a control current or a control voltage to the bias circuit **3**. The comparing section **6** compares a threshold voltage V_{ref} with a power supply voltage V_{cc} of a power supply terminal **11** connected to the output terminal of the specific transistor **Tr2A**. The signal input terminal **13** is connected to the input terminal of the specific transistor **Tr2A**, and a radio-frequency signal is input to the signal input terminal **13**. The attenuation circuit **4** is connected in a signal path **P1** between the signal input terminal **13** and the antenna terminal **105** and is capable of attenuating a radio-frequency signal. The control unit **55** changes an attenuation of the attenuation circuit **4** in accordance with a compared result of the comparing section **6**.

(61) In the radio-frequency circuit **100** according to the first embodiment, the control unit **55** changes the attenuation of the attenuation circuit **4** in accordance with a compared result of the comparing section **6**. Therefore, when, for example, the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , it is possible to attenuate a radio-frequency signal. In other words, with the radio-frequency circuit **100** according to the first embodiment, the output power of the amplifier circuit **2** is further accurately limited.

(62) In the radio-frequency circuit **100** according to the first embodiment, the attenuation circuit **4** is made up of a resistive element **41** and a switch element **42**. Therefore, the small-sized attenuation circuit **4** is implemented with a simple configuration.

(63) A radio-frequency module including the radio-frequency circuit **100** includes a plurality of circuit elements (a GaAs IC chip and an Si IC chip of the power amplifier circuit **1**, the first switch

102, the filter **103**, the second switch **104**, and the like) included in the radio-frequency circuit **100**, and a mounting substrate on which the plurality of circuit elements is mounted. In the radio-frequency module, to reduce the size of the radio-frequency module, a reduction in the size of the filter **103** of which a footprint on the mounting substrate tends to increase, of the plurality of circuit elements, can be attempted. Here, the power handling capability of the filter **103** tends to decrease as the chip size of the filter **103** reduces. In the radio-frequency circuit **100** according to the embodiment, the output power of the amplifier circuit **2** can be accurately limited, so, for example, the degradation of the characteristics of the filter **103** can be suppressed, with the result that the design margin of the filter **103** is easily expanded.

(64) (2.2) Communication Device

(65) A communication device **300** according to the first embodiment includes the radio-frequency circuit **100** and a signal processing circuit **301**. The signal processing circuit **301** is connected to the radio-frequency circuit **100**.

(66) The communication device **300** according to the first embodiment includes the radio-frequency circuit **100**, so the communication device **300** is capable of further accurately limiting the output power of the amplifier circuit **2**.

(67) (3) Modifications

(68) Hereinafter, modifications of the first embodiment will be listed. The modifications described below are applicable in combination as needed.

(69) (3.1) First Modification

(70) In the radio-frequency circuit **100**, an attenuation circuit **4a** may be applied instead of the attenuation circuit **4**. Hereinafter, the attenuation circuit **4a** will be described with reference to FIG. **3B**.

(71) As shown in FIG. **3B**, the attenuation circuit **4a** includes a resistive element **41a** and a switch element **42a**. In the attenuation circuit **4a**, the resistive element **41a** and the switch element **42a** are connected in series such that the switch element **42a** is on the signal path **P1** side and the resistive element **41a** is on the ground side. In other words, the attenuation circuit **4a** is connected between the signal path **P1** and the ground.

(72) The control unit **55** changes the attenuation of the attenuation circuit **4a** in accordance with a compared result of the comparing section **6**. Specifically, the control unit **55** controls the switch element **42a** to an on state when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , and controls the switch element **42a** to an off state when the power supply voltage V_{cc} is lower than or equal to the threshold voltage V_{ref} . Thus, when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , the resistive element **41a** is connected to the signal path **P1**, with the result that a radio-frequency signal is attenuated.

(73) With the radio-frequency circuit **100** according to the first modification, the small-sized attenuation circuit **4a** is implemented with a simple configuration.

(74) In FIG. **3B**, the number of the resistive elements **41a** is one. Alternatively, the number of the resistive elements **41a** may be multiple. In this case, the plurality of resistive elements **41a** may be connected in series or may be connected in parallel or may be in a combination of series and parallel connections.

(75) (3.2) Second Modification

(76) In the radio-frequency circuit **100**, an attenuation circuit **4b** may be applied instead of the attenuation circuit **4**. Hereinafter, the attenuation circuit **4b** will be described with reference to FIG. **3C**.

(77) As shown in FIG. **3C**, the attenuation circuit **4b** includes a plurality of rectifier elements **41b** and a switch element **42b**. Each of the plurality of rectifier elements **41b** is, for example, a diode. In the attenuation circuit **4b**, the plurality of rectifier elements **41b** is connected in series with each other. In the attenuation circuit **4b**, the plurality of rectifier elements **41b** and the switch element **42b** are connected in series such that the switch element **42b** is on the signal path **P1** side and the

plurality of rectifier elements **41b** is on the ground side. In other words, the attenuation circuit **4b** is connected between the signal path **P1** and the ground.

(78) The control unit **55** changes the attenuation of the attenuation circuit **4b** in accordance with a compared result of the comparing section **6**. Specifically, the control unit **55** controls the switch element **42b** to an on state when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , and controls the switch element **42b** to an off state when the power supply voltage V_{cc} is lower than or equal to the threshold voltage V_{ref} . Thus, when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , the plurality of rectifier elements **41b** is connected to the signal path **P1**, with the result that a radio-frequency signal is attenuated.

(79) With the radio-frequency circuit **100** according to the second modification, the small-sized attenuation circuit **4b** is implemented with a simple configuration. In the radio-frequency circuit **100** according to the second modification, the rectifier elements **41b** are used, so the output power can be limited to lower than or equal to a certain value. In other words, with the radio-frequency circuit **100** according to the second modification, an increase in input current can be handled.

(80) In FIG. **3C**, the plurality of rectifier elements **41b** is connected in series. Alternatively, the plurality of rectifier elements **41b** may be connected in parallel or may be in a combination of series and parallel connections. The number of the rectifier elements **41b** is not limited to multiple. The number of the rectifier elements **41b** may be one.

(81) (3.3) Third Modification

(82) In the radio-frequency circuit **100**, an attenuation circuit **4c** may be applied instead of the attenuation circuit **4**. Hereinafter, the attenuation circuit **4c** will be described with reference to FIG. **3D**.

(83) As shown in FIG. **3D**, the attenuation circuit **4c** includes a T-shape circuit **41c** and a switch element **42c**. The T-shape circuit **41c** has two first resistive elements **411**, **412** and a second resistive element **413**. The two first resistive elements **411**, **412** are connected in series with each other and connected in series with the signal path **P1**. The second resistive element **413** is connected between a ground and a junction point between the two first resistive elements **411**, **412**. The switch element **42c** is connected in parallel with the two first resistive elements **411**, **412** connected in series with the signal path **P1** in the T-shape circuit **41c**. In other words, in the attenuation circuit **4c**, the switch element **42c** and the T-shape circuit **41c** are connected in parallel.

(84) The control unit **55** changes the attenuation of the attenuation circuit **4c** in accordance with a compared result of the comparing section **6**. Specifically, the control unit **55** controls the switch element **42c** to an off state when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , and controls the switch element **42c** to an on state when the power supply voltage V_{cc} is lower than or equal to the threshold voltage V_{ref} . Thus, when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , the T-shape circuit **41c** is connected to the signal path **P1**, with the result that a radio-frequency signal is attenuated.

(85) With the radio-frequency circuit **100** according to the third modification, fluctuations in the impedance are suppressed by connecting the T-shape circuit **41c** to the signal path **P1**, so fluctuations in RF characteristics are suppressed.

(86) In FIG. **3D**, the number of the second resistive elements **413** is one. Alternatively, the number of the second resistive elements **413** may be multiple. In the third modification, the number of the parallel circuits made up of the switch element **42c** and the T-shape circuit **41c** is one. Alternatively, two or more parallel circuits may be connected to the signal path **P1**.

(87) (3.4) Fourth Modification

(88) In the radio-frequency circuit **100**, an attenuation circuit **4d** may be applied instead of the attenuation circuit **4**. Hereinafter, the attenuation circuit **4d** will be described with reference to FIG. **3E**.

(89) As shown in FIG. **3E**, the attenuation circuit **4d** includes a π -shape circuit **41d** and a switch element **42d**. The π -shape circuit **41d** has a first resistive elements **414** and two second resistive

elements **415**, **416**. The first resistive element **414** is connected in series with the signal path **P1**. Each of the two second resistive elements **415**, **416** is connected between the signal path **P1** and the ground. The switch element **42d** is connected in parallel with the first resistive element **414** connected in series with the signal path **P1** in the π -shape circuit **41d**. In other words, in the attenuation circuit **4d**, the switch element **42d** and the π -shape circuit **41d** are connected in parallel. (90) The control unit **55** changes the attenuation of the attenuation circuit **4c** in accordance with a compared result of the comparing section **6**. Specifically, the control unit **55** controls the switch element **42d** to an off state when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , and controls the switch element **42d** to an on state when the power supply voltage V_{cc} is lower than or equal to the threshold voltage V_{ref} . Thus, when the power supply voltage V_{cc} exceeds the threshold voltage V_{ref} , the π -shape circuit **41d** is connected to the signal path **P1**, with the result that a radio-frequency signal is attenuated.

(91) In FIG. **3E**, the number of the first resistive elements **414** is one. Alternatively, the number of the first resistive elements **414** may be multiple. Each of the number of the second resistive elements **415** and the number of the second resistive elements **416** is one. Alternatively, each of the number of the second resistive elements **415** and the number of the second resistive elements **416** may be multiple. Furthermore, in the fourth modification, the number of the parallel circuits made up of the switch element **42d** and the π -shape circuit **41d** is one. Alternatively, two or more parallel circuits may be connected to the signal path **P1**.

(92) (3.5) Fifth Modification

(93) In the radio-frequency circuit **100**, an attenuation circuit **4e** may be applied instead of the attenuation circuit **4**. Hereinafter, the attenuation circuit **4e** will be described with reference to FIG. **4A**.

(94) As shown in FIG. **4A**, the attenuation circuit **4e** includes a plurality of (two in the illustrated example) resistive elements **41e** and a plurality of (two in the illustrated example) switch elements **42e**. The plurality of resistive elements **41e** and the plurality of switch elements **42e** are provided in a one-to-one correspondence with each other. The plurality of resistive elements **41e** is connected in series with each other and connected in series with the signal path **P1**. Each of the plurality of switch elements **42e** is connected in parallel with an associated one of the plurality of resistive elements **41e**. In other words, the attenuation circuit **4e** has two parallel circuits each made up of one resistive element **41e** and one switch element **42e**. The two parallel circuits are connected in series with each other.

(95) In the radio-frequency circuit **100** according to the fifth modification, the amplifier circuit **2**, as in the case of the amplifier circuit **2** of the radio-frequency circuit **100** according to the first embodiment, is capable of operating in each of a plurality of (for example, two) communication bands (a first communication band and a second communication band) different from each other. Therefore, the specific transistor **Tr2A** is capable of operating in each of the plurality of communication bands different from each other. In the radio-frequency circuit **100** according to the fifth modification, the control unit **55** changes the attenuation of the attenuation circuit **4e** in accordance with, of the plurality of communication bands, the communication band in which the specific transistor **Tr2A** operates. The control unit **55** can acquire information related to the communication band in which the specific transistor **Tr2A** operates from the signal processing circuit **301**. The control unit **55** can change the attenuation of the attenuation circuit **4e** in accordance with, for example, a compared result of the comparing section **6**, and a control signal from the signal processing circuit **301**.

(96) In the radio-frequency circuit **100** according to the fifth modification, the control unit **55** sets all the plurality of switch elements **42e** to an on state when, for example, the power supply voltage V_{cc} is lower than the threshold voltage V_{ref} in accordance with a compared result of the comparing section **6**. When the control unit **55** is configured to set the attenuation of the attenuation circuit **4e** to a maximum value in accordance with the first communication band when, for example, the

power supply voltage V_{cc} is higher than the threshold voltage V_{ref} in accordance with a compared result of the comparing section 6, the control unit 55 sets all the plurality of switch elements 42e to an off state. When the control unit 55 is configured to set the attenuation of the attenuation circuit 4e to a minimum value in accordance with the second communication band when, for example, the power supply voltage V_{cc} is higher than the threshold voltage V_{ref} in accordance with a compared result of the comparing section 6, the control unit 55 sets one of the plurality of switch elements 42e to an off state and sets the remainder to an on state.

(97) With the radio-frequency circuit 100 according to the fifth modification, it is possible to further accurately limit the output power of the amplifier circuit 2 in any of the plurality of communication bands in which the specific transistor Tr2A operates.

(98) In FIG. 4A, each of the number of the resistive elements 41e and the number of the switch elements 42e is two; however, the configuration is not limited thereto. Alternatively, each of the number of the resistive elements 41e and the number of the switch elements 42e may be three or more. In other words, three or more parallel circuits each made up the resistive element 41e and the switch element 42e may be connected in series.

(99) (3.6) Sixth Modification

(100) In the radio-frequency circuit 100, an attenuation circuit 4f may be applied instead of the attenuation circuit 4e. Hereinafter, the attenuation circuit 4f will be described with reference to FIG. 4B.

(101) As shown in FIG. 4B, the attenuation circuit 4f includes a plurality of (three in the illustrated example) resistive elements 41f and a plurality of (three in the illustrated example) switch elements 42f. The plurality of resistive elements 41f and the plurality of switch elements 42f are provided in a one-to-one correspondence with each other. Each of the plurality of resistive elements 41f is connected in series with an associated one of the plurality of switch elements 42f. In other words, the attenuation circuit 4f has a plurality of series circuits each made up of one resistive element 41f and one switch element 42f. Each of the plurality of series circuits is connected between the signal path P1 and the ground.

(102) With the radio-frequency circuit 100 according to the sixth modification, as in the case of the radio-frequency circuit 100 according to the fifth modification, it is possible to further accurately limit the output power of the amplifier circuit 2 in any one of the plurality of communication bands in which the specific transistor Tr2A operates.

(103) In FIG. 4B, each of the number of the resistive elements 41f and the number of the switch elements 42f is three; however, the configuration is not limited thereto. Each of the number of the resistive elements 41f and the number of the switch elements 42f may be two or may be four or more. In other words, the number of series circuits each made up of the resistive element 41e and the switch element 42e may be two or may be four or more.

(104) (3.7) Seventh Modification

(105) In the radio-frequency circuit 100, an attenuation circuit 4g may be applied instead of the attenuation circuit 4e. Hereinafter, the attenuation circuit 4g will be described with reference to FIG. 4C.

(106) As shown in FIG. 4C, the attenuation circuit 4g includes a plurality of rectifier elements 41g, a first switch element 42g, and a plurality of (three in the illustrated example) second switch elements 43g. In FIG. 4C, the plurality of rectifier elements 41g is divided into three groups each having a plurality of the rectifier elements 41g. In each group, the plurality of rectifier elements 41g is connected in series with each other. The plurality of groups is connected in parallel with one another. The plurality of second switch elements 43g and the plurality of groups are in a one-to-one correspondence with each other. Each of the plurality of second switch elements 43g is connected in parallel with an associated one of the plurality of groups. The first switch element 42g is connected in series with the plurality of groups connected in parallel with one another. As shown in FIG. 4C, the attenuation circuit 4g according to the seventh modification is connected between the

signal path P1 and the ground.

(107) With the radio-frequency circuit **100** according to the seventh modification, as in the case of the radio-frequency circuit **100** according to the fifth modification, it is possible to further accurately limit the output power of the amplifier circuit **2** in any of the plurality of communication bands in which the specific transistor Tr2A operates.

(108) In FIG. **4C**, the number of the second switch elements **43g** is three; however, the configuration is not limited thereto. The number of the second switch elements **43g** may be, for example, two or may be four or more. The plurality of rectifier elements **41g** that make up each group is connected in series. Alternatively, the plurality of rectifier elements **41g** may be connected in parallel or may be in a combination of series and parallel connections. Furthermore, the number of the rectifier elements **41g** that make up each group may be one.

(109) (3.8) Eighth Modification

(110) A radio-frequency circuit **100** according to an eighth modification has the same circuit configuration as the radio-frequency circuit **100** according to the first embodiment, so the description will be made with reference to FIGS. **1** and **2**.

(111) In the radio-frequency circuit **100** according to the eighth modification, the control unit **55** acquires information related to the rated power supply voltage of the power supply voltage Vcc applied to the power supply terminal **11** and changes the threshold voltage Vref in accordance with the information. The control unit **55** acquires information related to the rated power supply voltage of the power supply voltage Vcc from, for example, the signal processing circuit **301**.

(112) In the radio-frequency circuit **100** according to the eighth modification, the control unit **55**, for example, increases the threshold voltage Vref as the rated power supply voltage increases in accordance with information on the acquired rated power supply voltage.

(113) In the radio-frequency circuit **100** according to the eighth modification, when, for example, the specifications of the rated power supply voltage of the power supply voltage Vcc applied to the power supply terminal **11** are changed, the threshold voltage Vref can be changed in accordance with the rated power supply voltage.

(114) (3.9) Other Modifications

(115) For example, in the radio-frequency circuit **100**, the number of stages of the transistors Tr2 connected in the amplifier circuit **2** is two; however, the configuration is not limited thereto. The number of stages of the transistors Tr2 connected may be three or more. The amplifier circuit **2** is not limited to a configuration having a plurality of transistors Tr2 and may be the one having at least one transistor Tr2. In short, in the radio-frequency circuit **100**, the number of transistors Tr2 may be one. In this case, each of the number of the emitter followers **30** and the number of the bipolar transistors Tr3 in the bias circuit **3** may be one. In the radio-frequency circuit **100**, the first-stage transistor Tr21 is set as the specific transistor Tr2A; however, the configuration is not limited thereto. The final-stage transistor Tr22 may be set as the specific transistor Tr2A or each of the first-stage transistor Tr21 and the final-stage transistor Tr22 may be set as the specific transistor Tr2A. When the number of stages of the transistors Tr2 connected is three or more, the transistor Tr2 other than both the first-stage transistor Tr21 and the final-stage transistor Tr22 may be set as the specific transistor Tr2A.

(116) The transistors Tr2 in the amplifier circuit **2** are not limited to bipolar transistors and may be field effect transistors (FETs). When the transistors Tr2 are FETs, the gate, the drain, and the source are respectively an input terminal, an output terminal, and a common terminal.

(117) In the radio-frequency circuit **100**, for example, the first switch **102**, the filter **103**, and the second switch **104** are not indispensable component elements and are additional component elements.

(118) In the radio-frequency circuit **100**, the filter **103** is an acoustic wave filter that uses surface acoustic waves; however, the configuration is not limited thereto. The filter **103** may be, for example, an acoustic wave filter that uses boundary acoustic waves, plate waves, or the like.

(119) In the acoustic wave filter, each of the plurality of series arm resonators and the plurality of parallel arm resonators is not limited to an SAW resonator and may be, for example, a bulk acoustic wave (BAW) resonator.

(120) The radio-frequency circuit **100** may include a receiving circuit including a low-noise amplifier that amplifies a reception signal input from the antenna terminal **105** and a filter connected to the low-noise amplifier.

(121) The filter **103** is not limited to a transmission filter and may be a duplexer.

(122) In the radio-frequency circuit **100**, the first switch **102** and the second switch **104** each may be, for example, a switch IC that supports general purpose input/output (GPIO).

(123) A voltage input to the first input terminal of the comparator CP1 is not limited to the power supply voltage Vcc and may be, for example, a voltage into which the power supply voltage Vcc is divided by two resistances. In this case, the threshold voltage Vref input to the second input terminal of the comparator CP1 can be suppressed to a lower voltage.

Second Embodiment

(124) A radio-frequency circuit **100** according to a second embodiment will be described with reference to FIG. 5. Regarding the radio-frequency circuit **100** according to the second embodiment, like reference signs denote component elements similar to those of the radio-frequency circuit **100** according to the first embodiment, and the description is omitted.

(125) The radio-frequency circuit **100** according to the second embodiment differs from the radio-frequency circuit **100** according to the first embodiment in that the attenuation circuit **4** is provided on the input side of the first-stage amplifier **21**. In other words, in the radio-frequency circuit **100** according to the second embodiment, the attenuation circuit **4** is connected in the signal path P1 between the signal input terminal **13** and the matching circuit **23** connected to the input terminal of the first-stage amplifier **21**, as shown in FIG. 5.

(126) In the radio-frequency circuit **100** according to the second embodiment, the attenuation circuit **4** is provided on the input side of the first-stage amplifier **21**, so the input power input to the first-stage amplifier **21** can be limited, with the result that the degradation of the first-stage amplifier **21** can be suppressed. In the radio-frequency circuit **100** according to the second embodiment, as in the case of the radio-frequency circuit **100** according to the first embodiment, the output power of the amplifier circuit **2** is further accurately limited.

(127) The attenuation circuit **4** may be connected between the matching circuit **23** and the first-stage amplifier **21**.

(128) In the second embodiment as well, any one of the attenuation circuits **4a** to **4g** may be applied instead of the attenuation circuit **4**.

Third Embodiment

(129) A radio-frequency circuit **100** according to a third embodiment will be described with reference to FIG. 6. Regarding the radio-frequency circuit **100** according to the third embodiment, like reference signs denote component elements similar to those of the radio-frequency circuit **100** according to the first embodiment, and the description is omitted.

(130) The radio-frequency circuit **100** according to the third embodiment differs from the radio-frequency circuit **100** according to the first embodiment in that the attenuation circuit **4** is provided on the output side of the final-stage amplifier **22**. In other words, in the radio-frequency circuit **100** according to the third embodiment, the attenuation circuit **4** is connected in the signal path P1 between the first switch **102** and the output matching circuit **101** connected to the output terminal of the final-stage amplifier **22**, as shown in FIG. 6.

(131) In the radio-frequency circuit **100** according to the third embodiment, the attenuation circuit **4** is provided on the output side of the final-stage amplifier **22**, so the input power input to the first switch **102** can be limited, with the result that the degradation of the first switch **102** can be suppressed. In the radio-frequency circuit **100** according to the third embodiment, as in the case of the radio-frequency circuit **100** according to the first embodiment, the output power of the

amplifier circuit **2** can be further accurately limited

(132) The attenuation circuit **4** may be connected between the final-stage amplifier **22** and the output matching circuit **101** or may be connected between the first switch **102** and the filter **103**.

(133) In the third embodiment as well, any one of the attenuation circuits **4a** to **4g** may be applied instead of the attenuation circuit **4**.

Fourth Embodiment

(134) A radio-frequency circuit **100** according to a fourth embodiment will be described with reference to FIG. 7. Regarding the radio-frequency circuit **100** according to the fourth embodiment, like reference signs denote component elements similar to those of the radio-frequency circuit **100** according to the first embodiment, and the description is omitted.

(135) The radio-frequency circuit **100** according to the fourth embodiment differs from the radio-frequency circuit **100** according to the first embodiment in that the attenuation circuit **4** is connected to the antenna terminal **105**. In other words, in the radio-frequency circuit **100** according to the fourth embodiment, the attenuation circuit **4** is connected in the signal path **P1** between the second switch **104** and the antenna terminal **105**, as shown in FIG. 7.

(136) In the radio-frequency circuit **100** according to the fourth embodiment, the attenuation circuit **4** is provided between the second switch **104** and the antenna terminal **105**, so the influence of the antenna **310** in the event of fluctuations in load is reduced. In the radio-frequency circuit **100** according to the fourth embodiment, as in the case of the radio-frequency circuit **100** according to the first embodiment, the output power of the amplifier circuit **2** can be further accurately limited.

(137) In the fourth embodiment as well, any one of the attenuation circuits **4a** to **4g** may be applied instead of the attenuation circuit **4**.

Aspects

(138) The following aspects are disclosed in the specification.

(139) A radio-frequency circuit (**100**) according to a first aspect includes an amplifier circuit (**2**), a bias circuit (**3**), a bias control circuit (**54**), a comparing section (**6**), a signal input terminal (**13**), an antenna terminal (**105**), an attenuation circuit (**4**; **4a** to **4g**), and a control unit (**55**). The amplifier circuit (**2**) includes a specific transistor (Tr2A). The specific transistor (Tr2A) has an input terminal and an output terminal. The specific transistor (Tr2A) amplifies a radio-frequency signal input to the input terminal and outputs the radio-frequency signal from the output terminal. The bias circuit (**3**) supplies a bias current or a bias voltage to the input terminal of the specific transistor (Tr2A). The bias control circuit (**54**) supplies a control current (I_{cont}) or a control voltage to the bias circuit (**3**). The comparing section (**6**) compares a threshold voltage (V_{ref}) with a power supply voltage (V_{cc}) of a power supply terminal (**11**) connected to the output terminal of the specific transistor (Tr2A). The signal input terminal (**13**) is connected to the input terminal of the specific transistor (Tr2A), and a radio-frequency signal is input to the signal input terminal (**13**). The attenuation circuit (**4**; **4a** to **4g**) is connected in a signal path (**P1**) between the signal input terminal (**13**) and the antenna terminal (**105**) and is capable of attenuating a radio-frequency signal. The control unit (**55**) changes an attenuation of the attenuation circuit (**4**; **4a** to **4g**) in accordance with a compared result of the comparing section (**6**).

(140) According to this aspect, the output power of the amplifier circuit (**2**) is further accurately limited.

(141) In the radio-frequency circuit (**100**) according to a second aspect, in the first aspect, the amplifier circuit (**2**) has a first-stage transistor (Tr21) serving as the specific transistor (Tr2A), and a final-stage transistor (Tr22). The final-stage transistor (Tr22) is connected at a subsequent stage to the first-stage transistor (Tr21). The attenuation circuit (**4**; **4a** to **4g**) is connected in the signal path (**P1**) between the first-stage transistor (Tr21) and the final-stage transistor (Tr22).

(142) According to this aspect, the output power of the amplifier circuit (**2**) is further accurately limited.

(143) In the radio-frequency circuit (**100**) according to a third aspect, in the first aspect, the

attenuation circuit (**4**; **4a** to **4g**) is connected in the signal path (**P1**) between the signal input terminal (**13**) and the specific transistor (Tr2A).

(144) According to this aspect, the output power of the amplifier circuit (**2**) is further accurately limited.

(145) The radio-frequency circuit (**100**) according to a fourth aspect, in the first aspect, further includes a switch (**102**). The switch (**102**) is connected between the specific transistor (Tr2A) and the antenna terminal (**105**). The attenuation circuit (**4**; **4a** to **4g**) is connected in the signal path (**P1**) between the specific transistor (Tr2A) and the switch (**102**).

(146) According to this aspect, the output power of the amplifier circuit (**2**) is further accurately limited.

(147) The radio-frequency circuit (**100**) according to a fifth aspect, in the first aspect, further includes a switch (**104**). The switch (**104**) is connected to the antenna terminal (**105**). The attenuation circuit (**4**; **4a** to **4g**) is connected in the signal path (**P1**) between the switch (**104**) and the antenna terminal (**105**).

(148) According to this aspect, the output power of the amplifier circuit (**2**) is further accurately limited.

(149) In the radio-frequency circuit (**100**) according to a sixth aspect, in any one of the first to fifth aspects, the attenuation circuit (**4**) includes a switch element (**42**) and at least one resistive element (**41**). The switch element (**42**) is turned on or off in accordance with a compared result of the comparing section (**6**). The at least one resistive element (**41**) is connected in parallel with the switch element (**42**). The at least one resistive element (**41**) is connected in series with the signal path (**P1**).

(150) According to this aspect, the small-sized attenuation circuit (**4**) is implemented with a simple configuration.

(151) In the radio-frequency circuit (**100**) according to a seventh aspect, in any one of the first to fifth aspects, the attenuation circuit (**4a**) includes a switch element (**42a**) and at least one resistive element (**41a**). The switch element (**42a**) is turned on or off in accordance with a compared result of the comparing section (**6**). The at least one resistive element (**41a**) is connected in series with the switch element (**42a**). The attenuation circuit (**4a**) is connected between the signal path (**P1**) and a ground.

(152) According to this aspect, the small-sized attenuation circuit (**4a**) is implemented with a simple configuration.

(153) In the radio-frequency circuit (**100**) according to an eighth aspect, in any one of the first to fifth aspects, the attenuation circuit (**4b**) includes a switch element (**42b**) and at least one rectifier element (**41b**). The switch element (**42b**) is turned on or off in accordance with a compared result of the comparing section (**6**). The at least one rectifier element (**41b**) is connected in series with the switch element (**42b**). The attenuation circuit (**4b**) is connected between the signal path (**P1**) and a ground.

(154) According to this aspect, the small-sized attenuation circuit (**4b**) is implemented with a simple configuration.

(155) In the radio-frequency circuit (**100**) according to a ninth aspect, in any one of the first to fifth aspects, the attenuation circuit (**4c**) includes a switch element (**42c**) and a T-shape circuit (**41c**). The switch element (**42c**) is connected in series with the signal path (**P1**) and is turned on or off in accordance with a compared result of the comparing section (**6**). The T-shape circuit (**41c**) is connected in parallel with the switch element (**42c**). The T-shape circuit (**41c**) has two first resistive elements (**411**, **412**) and a second resistive element (**413**). The two resistive elements (**411**, **412**) are connected in parallel with the switch element (**42c**). The second resistive element (**413**) is connected between a ground and a junction point between the two first resistive elements (**411**, **412**).

(156) According to this aspect, variations in RF characteristics are suppressed.

(157) In the radio-frequency circuit (100) according to a tenth aspect, in any one of the first to ninth aspects, the specific transistor (Tr2A) is capable of operating in each of a plurality of communication bands different from each other. The control unit (55) changes the attenuation of the attenuation circuit (4e; 4f; 4g) in accordance with, of the plurality of communication bands, the communication band in which the specific transistor (Tr2A) operates.

(158) According to this aspect, the output power of the amplifier circuit (2) can be further accurately limited in accordance with the communication band in which the specific transistor (Tr2A) operates.

(159) In the radio-frequency circuit (100) according to an eleventh aspect, in any one of the first to ninth aspects, the control unit (55) acquires information related to a rated power supply voltage of the power supply voltage (Vcc) applied to the power supply terminal (11) and changes the threshold voltage (Vref) in accordance with the information.

(160) According to this aspect, when, for example, the specifications of the rated power supply voltage of the power supply voltage (Vcc) applied to the power supply terminal (11) are changed, the threshold voltage (Vref) can be changed in accordance with the rated power supply voltage.

(161) In the radio-frequency circuit (100) according to a twelfth aspect, in any one of the first to eleventh aspects, the comparing section (6) includes a comparator (CP1) that compares the threshold voltage (Vref) with the power supply voltage (Vcc) of the power supply terminal (11). The control unit (55) changes the attenuation of the attenuation circuit (4; 4a to 4g) in accordance with an output of the comparator (CP1).

(162) According to this aspect, in comparison with the case where the power supply voltage (Vcc) of the power supply terminal (11) is divided in accordance with resistances by a resistance voltage circuit and is compared with the threshold voltage, a voltage drop between the power supply terminal (11) and the output terminal of the specific transistor (Tr2A) is reduced, so a decrease in the output power of the amplifier circuit (2) and a decrease in operation efficiency can be suppressed.

(163) A communication device (300) according to a thirteenth aspect includes the radio-frequency circuit (100) that is any one of the first to twelfth aspects, and a signal processing circuit (301). The signal processing circuit (301) is connected to the radio-frequency circuit (100).

(164) According to this aspect, the output power of the amplifier circuit (2) is further accurately limited.

REFERENCE SIGNS LIST

(165) 1 power amplifier circuit 2 amplifier circuit 3 bias circuit 4, 4a to 4g attenuation circuit 5 control circuit 6 comparing section 11 power supply terminal 12 battery terminal 13 signal input terminal 20 amplifier 21 first-stage amplifier (specific transistor, first-stage transistor) 22 final-stage amplifier (final-stage transistor) 23, 24, 25 matching circuit 27 input terminal 28 output terminal 30 emitter follower 31 first emitter follower 32 second emitter follower 41, 41a, 41e, 41f resistive element 41b, 41g rectifier element 41c T-shape circuit 41d π -shape circuit 42, 42a to 42g, 43g switch element 52 current source 54 bias control circuit 55 control unit 100 radio-frequency circuit 101 output matching circuit 102 first switch (switch) 103 filter 104 second switch (switch) 105 antenna terminal 300 communication device 301 signal processing circuit 302 RF signal processing circuit 303 baseband signal processing circuit 310 antenna 330 power management IC 411, 412, 414 first resistive element 413, 415, 416 second resistive element CP1 comparator D311, D312, D321, D322 diode Ice current Icont control current N1 node P1 signal path R31, R32 resistance Tr2 transistor Tr21 first-stage transistor Tr22 final-stage transistor Tr3 bipolar transistor Tr31 first bipolar transistor Tr32 second bipolar transistor Vcc power supply voltage Vref threshold voltage

Claims

1. A radio-frequency circuit comprising: an amplifier circuit comprising a specific transistor that has an input terminal and an output terminal, and that is configured to amplify a radio-frequency signal input to the input terminal and to output an amplified radio-frequency signal from the output terminal; a bias circuit configured to supply a bias current or a bias voltage to the input terminal of the specific transistor; a bias control circuit configured to supply a control current or a control voltage to the bias circuit; comparing circuitry configured to compare a threshold voltage with a power supply voltage of a power supply terminal connected to the output terminal of the specific transistor; a signal input terminal connected to the input terminal of the specific transistor, and to which the radio-frequency signal is input; an antenna terminal; an attenuation circuit connected in a signal path between the signal input terminal and the antenna terminal, and configured to attenuate the radio-frequency signal; and control circuitry configured to change an attenuation of the attenuation circuit in accordance with a result of the comparing section.
2. The radio-frequency circuit according to claim 1, wherein the amplifier circuit comprises: the specific transistor, the specific transistor being a first-stage transistor of the amplifier circuit, and a final-stage transistor connected at a stage subsequent to the first-stage transistor, and wherein the attenuation circuit is connected in the signal path between the first-stage transistor and the final-stage transistor.
3. The radio-frequency circuit according to claim 1, wherein the attenuation circuit is connected in the signal path between the signal input terminal and the specific transistor.
4. The radio-frequency circuit according to claim 1, further comprising a switch connected between the specific transistor and the antenna terminal, wherein the attenuation circuit is connected in the signal path between the specific transistor and the switch.
5. The radio-frequency circuit according to claim 1, further comprising a switch connected to the antenna terminal, wherein the attenuation circuit is connected in the signal path between the switch and the antenna terminal.
6. The radio-frequency circuit according to claim 1, wherein the attenuation circuit comprises: a switch that is configured to turn on or off in accordance with the result of the comparing circuitry, and at least one resistive circuit element connected in parallel with the switch, and wherein the at least one resistive circuit element is connected in series with the signal path.
7. The radio-frequency circuit according to claim 1, wherein the attenuation circuit comprises: a switch that is configured to turn on or off in accordance with the result of the comparing circuitry, and at least one resistive circuit element connected in series with the switch, and wherein the attenuation circuit is connected between the signal path and ground.
8. The radio-frequency circuit according to claim 1, wherein the attenuation circuit comprises: a switch that is configured to turn on or off in accordance with the result of the comparing circuitry, and at least one rectifier circuit element connected in series with the switch, and wherein the attenuation circuit is connected between the signal path and ground.
9. The radio-frequency circuit according to claim 1, wherein the attenuation circuit comprises: a switch that is connected in series with the signal path, and that is configured to turn on or off in accordance with the result of the comparing circuitry, and a T-shape circuit connected in parallel with the switch, and wherein the T-shape circuit comprises: two first resistive circuit elements connected in parallel with the switch, and a second resistive circuit element connected between ground and a node between the two first resistive circuit elements.
10. The radio-frequency circuit according to claim 1, wherein the specific transistor is configured to operate in each of a plurality of communication bands different from each other, and the control circuitry is configured to change the attenuation of the attenuation circuit in accordance with the communication band of the plurality of communication bands in which the specific transistor operates.
11. The radio-frequency circuit according to claim 1, wherein the control circuitry is configured to

acquire information related to a rated power supply voltage of the power supply voltage applied to the power supply terminal, and to change the threshold voltage in accordance with the information.

12. The radio-frequency circuit according to claim 1, wherein the comparing circuitry comprises a comparator configured to compare the power supply voltage of the power supply terminal with the threshold voltage, and wherein the control circuitry is configured to change the attenuation of the attenuation circuit in accordance with an output of the comparator.

13. A communication device comprising: the radio-frequency circuit according to claim 1; and a signal processing circuit connected to the radio-frequency circuit.
